

Measuring Training-Free Tool Routing for Multimodal Agents

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Abstract

Multimodal agents increasingly have access to OCR, crop/zoom, code execution, search, and other tools, but a central evaluation question remains under-specified: when is a tool call worth its added cost, latency, and failure surface? This paper studies that question as answer-level admission control rather than as a new trained-agent recipe. We evaluate UtilityRouter, a non-oracle router that uses observable task text, image-presence metadata, answer type, benchmark metadata, and first-pass uncertainty before final scoring, comparing it with direct-answer, always-tool, and single-tool controls on selected official slices from MathVista, OCRBench, and SEED-Bench-2-Plus. All conditions use the hosted gpt-5-mini multimodal backbone under a one- or two-call answer budget, and the 480-row comparison keeps task counts, metrics, answer-call budgets, and tool-call rates visible rather than hiding them in an aggregate average. Across 480 scored rows, UtilityRouter ties direct answering on MathVista (20/30) and SEED-Bench-2-Plus (22/30), and is higher by one task on OCRBench (27/30 versus 26/30), with no paired comparison reaching significance. The resulting contribution is a calibrated protocol and failure analysis: tool utility is family-specific, non-oracle routing decisions can be audited before scoring, and selected-slice evidence can show when a router should call a tool, abstain, or defer to a simpler baseline.

1 Introduction

Multimodal language agents are commonly framed as systems that can decide when to reason, observe, and act with external tools. Classic tool-use and reasoning-action work established modular acting loops and tool calls for language agents (Karpas et al., 2022; Yao et al., 2022; Schick et al., 2023), while API-oriented systems and benchmarks made tool selection a concrete interface problem rather than only a prompting pattern (Qin et al., 2023; Li

et al., 2023b; Patil et al., 2023; Shen et al., 2023). In parallel, multimodal and web-agent benchmarks have made visual grounding, page interaction, and information retrieval central to agent evaluation (Zhou et al., 2023; Koh et al., 2024; Tian et al., 2024; Mialon et al., 2023). These threads leave an earlier decision under-measured: before an answer is scored, the agent must decide whether the current instance warrants OCR, crop/zoom, code execution, search, or no tool at all.

This paper treats that decision as answer-level admission control. The gap is methodological rather than another tool inventory: a router should be evaluated against direct answering, forced-tool controls, and single-tool policies under the same task slice, model snapshot, and scoring rule. Benchmarks such as MathVista, OCRBench, and SEED-Bench-2-Plus stress complementary parts of this admission problem—visual mathematical reasoning, text-rich image understanding, and text-rich visual comprehension (Lu et al., 2023a; Liu et al., 2023c; Li et al., 2024). Multimodal search and browsing benchmarks motivate the broader orchestration setting (Jiang et al., 2024; Koh et al., 2024; Wu et al., 2025), but this draft keeps those settings as motivation because their service and baseline dependencies are not part of the terminal evaluated evidence matrix.

We evaluate UtilityRouter, a training-free utility router whose decision rule is fixed before scoring. It converts observable question text, benchmark metadata, image-presence metadata, answer type, and a first-pass uncertainty signal into OCR/search/math/web flags, then admits tools through a small deterministic rule table. The terminal matrix contains 480 scored rows over 90 selected official tasks: 30 tasks each from MathVista, OCRBench, and SEED-Bench-2-Plus, with five or six conditions per family, all answered by the same hosted gpt-5-mini multimodal backbone.

The result separates three claims that are of-

ten conflated. UtilityRouter ties direct answering on MathVista (20/30) and SEED-Bench-2-Plus (22/30), is one selected task higher on OCR-Bench (27/30 versus 26/30), and loses to stronger crop/code controls on MathVista. These counts support a positive but narrow contribution: a non-oracle routing protocol with explicit direct, always-tool, and single-tool controls; an auditable admission rule whose inputs are inspectable before scoring; and a cross-family failure analysis showing when current training-free cues abstain, help weakly, or miss a useful tool path.

This framing is also motivated by a practical mismatch between benchmark design and tool admission. Many recent agent benchmarks evaluate the completed answer, trajectory, or browsing outcome, while the routing decision happens earlier and under less information. High-resolution visual suites, multimodal memory tasks, and realistic visual-agent benchmarks make this mismatch sharper because the agent may need to decide whether to zoom, extract text, call a detector, or defer to direct visual reasoning before it knows whether the action will pay off (Zhang et al., 2024; Huang et al., 2026; Su et al., 2026). The paper therefore treats routing as an admission-control layer whose quality should be measured against explicit controls, not as an implicit property of a larger agent loop.

The negative parts of the result are part of the contribution. If UtilityRouter had been reported only as an aggregate average, the MathVista failure would be easy to hide: the router ties direct answer, but two single-tool controls are higher. Conversely, OCRBench shows a weak positive router signal even though always-OCR does not dominate direct answering. These opposite regimes are exactly why the paper keeps direct, always-tool, and single-tool controls in the same table and avoids replacing them with a single headline score.

2 Related Work

Tool-using agents and routing. MRKL systems, ReAct, and Toolformer motivate modular tool access, reasoning-action loops, and learned or induced tool use (Karpas et al., 2022; Yao et al., 2022; Schick et al., 2023). Chameleon and ViperGPT further show how compositional and programmatic visual reasoning can use external modules or code execution (Lu et al., 2023b; Surís et al., 2023). These systems motivate the available tool set in this project, but this draft studies a different object: a

training-free admission policy that decides whether a tool call should happen under an answer-level evaluation budget.

ToolLLM, API-Bank, Gorilla, and HuggingGPT broaden the same design space toward large API inventories, API-use benchmarks, and delegation to external model or service hubs (Qin et al., 2023; Li et al., 2023b; Patil et al., 2023; Shen et al., 2023). They make the routing problem more explicit: once the action space contains many possible tools, the system needs an admission boundary that can decide when no tool is preferable to a poorly chosen one. Reflexion and Self-Refine show that feedback loops can improve agent behavior through verbal or iterative revision (Shinn et al., 2023; Madaan et al., 2023). Those methods are complementary to this paper because they change the policy or reasoning process after feedback, while the present protocol freezes a training-free route decision and asks whether the admitted tool call improves the final answer on the current item.

Multimodal agent benchmarks. VisualWebArena, MMInA, GAIA, AgentBench, and WebArena show that realistic agent tasks require visual, textual, and sometimes web-based observations (Koh et al., 2024; Tian et al., 2024; Mialon et al., 2023; Liu et al., 2023b; Zhou et al., 2023). Web-focused and operating-system agent benchmarks push the same issue toward longer interaction horizons, where a system must decide when to inspect a page, click, search, or use a computer interface rather than answer from the current observation (Deng et al., 2023; He et al., 2024; Xie et al., 2024). Recent multimodal search-agent papers further raise the frontier for search-heavy tasks (Jiang et al., 2024; Wu et al., 2025; Yan et al., 2026; Wang et al., 2026). Those papers motivate search and browser routing as long-term targets, but this draft does not use them as final evaluated evidence. The executed final matrix is limited to MathVista, OCRBench, and SEED-Bench-2-Plus. This distinction matters because browsing and search agents often combine retrieval, page interaction, memory, and answer synthesis in one evaluation loop. Benchmarks such as MM-BrowseComp, MME-RealWorld, M³Exam, and AgentVista broaden the domain pressure toward realistic visual interaction and longer-horizon context (Li et al., 2025; Zhang et al., 2024; Huang et al., 2026; Su et al., 2026). They are valuable frontier signals for future work, but the present claim cannot borrow their diffi-

culty without executing comparable final conditions. The paper therefore cites them as motivation for why routing matters while keeping the reported evidence on the three completed families.

Text-rich and visual reasoning evaluations. MathVista evaluates mathematical reasoning in visual contexts (Lu et al., 2023a); OCRBench isolates OCR-heavy capabilities in large multimodal models (Liu et al., 2023c); and SEED-Bench-2-Plus stresses text-rich visual comprehension (Li et al., 2024). These families are useful for a first routing-utility study because a direct-answer baseline, an OCR/crop/code tool policy, and a router can all be scored on the same selected tasks. The remaining gap is that benchmark-level aggregate scores do not directly answer whether the agent should call a tool on a given row. General multimodal model evaluations and model-building work such as MM-Vet, LLaVA, and BLIP-2 explain why direct multimodal answering is a strong baseline rather than a weak strawman (Yu et al., 2023; Liu et al., 2023a; Li et al., 2023a). Broad expert-level multimodal benchmarks such as MMMU reinforce the same point: many evaluation rows test integrated perception and reasoning rather than a single isolated tool call (Yue et al., 2023). Modern multimodal backends often solve many rows without auxiliary tools, so a router must beat or justify the cost of that direct path. This is one reason the present paper keeps direct answer in every family and treats tied accuracy with extra calls as a diagnostic result rather than as a win.

Over-privileged tool access. The routing problem is closely related to the recent observation that agents may receive more tool privilege than a task requires (Yang et al., 2026). For multimodal agents, over-privilege can mean calling OCR on a row whose answer is already visually obvious, invoking crop/zoom when the question is text-only, or escalating to browser/search behavior when the final evidence does not require it. Our setting is narrower than a general privilege-selection study, but it provides a measurable answer-level slice: we can ask whether the router’s admitted tool calls improve the official answer score, tie direct answering at higher cost, or miss a single-tool condition that would have been better.

3 Training-Free Utility Routing Protocol

UtilityRouter is a training-free controller for answer-level multimodal tasks. For each task, the system first builds an observable task view. The allowed fields include visible question text, benchmark family, task type or answer type, image-presence metadata, and other metadata that a deployed agent could inspect before scoring. The protocol explicitly excludes gold answers, evaluator outputs, reference answers, post-hoc correctness, and any exact lookahead over which condition will be correct.

The controller then chooses among a direct answer path and tool-mediated paths. In the final executed matrix, the implemented tools are OCR-style observation, crop/zoom-style visual access, and code-style reasoning for the families where those controls were provisioned. Search and browser actions are part of the project direction, but not part of the terminal answer-level evidence. The paper-facing comparisons are direct answering, always-tool baselines, single-tool routers, and UtilityRouter.

The route decision is framed as admission control rather than tool ranking over a fully known outcome table. For a row to be eligible for a tool path, the controller may inspect the family name, question form, answer type, lightweight image/task metadata, and a first-pass uncertainty signal. It may not inspect the gold answer, scorer output, or later condition-level success labels. This restriction separates the evaluated policy from an oracle selector. It also explains why the paper reports single-tool controls separately: if a single-tool control is stronger than UtilityRouter, the router failed to admit an available useful path rather than being denied access to that path.

The distinction between available and paper-facing tools is enforced at two levels. First, every result row is grouped by benchmark family and method condition, so the table exposes whether a claimed tool was actually executed. Second, the paper explicitly scopes search/browser utility and trained-search competitiveness out of the executed result claims, so those motivations cannot silently enter the abstract or results discussion. This is important for a training-free paper: the method may be designed for a broader tool menu, but only executed answer-level conditions can support final claims.

The routing policy is deliberately simple. It uses

observable features and a first-pass uncertainty signal from the same evaluated hosted multimodal backend, then applies fixed rules for whether to answer directly or request a tool observation. This design choice is not meant to outperform trained search or browser agents. It is meant to make the admission boundary inspectable: every final row records the observable task view, observable features, route decision, tool calls, answer, and score after prediction.

The implemented rule table is as follows. First, the task view is sanitized to visible fields such as benchmark family, question, choices, question/answer type, metadata, and image-presence metadata. The router derives four Boolean cues: OCR-like text if the visible prompt or metadata mentions reading, text, words, tables, or documents; search-like text if the task is a search family or asks who/when/where, release-date, conference, or search-style questions; math-like text if the family is MathVista or the prompt mentions calculation, distance, equation, value, or similar arithmetic cues; and web-like text for browser-family rows. It also marks a first-pass answer as uncertain when the first response contains phrases such as “not sure,” “cannot determine,” “unknown,” “need,” “search,” or “image.” Given those cues, UtilityRouter admits search for search-like rows when the first pass is uncertain or the family is a search family, admits OCR for OCR-like rows when the benchmark item includes an image or the first pass is uncertain, admits code for math-like rows whose answer type is integer or float, and admits crop/zoom when the benchmark item includes an image and the first pass is uncertain. The final selected-slice matrix exercises the OCR, crop/zoom, and code branches; search and browser branches remain scoped out of the paper-facing results.

The protocol therefore has two audit questions for every family. The first is an answer-quality question: does the routed condition score higher, tie, or fall below direct answering and the strongest single-tool control? The second is a mechanism question: when the routed condition ties direct answer, did it avoid unnecessary tools, or did it spend extra calls without changing the answer? The current selected-slice matrix answers these questions differently across the three families. MathVista is an admission failure because stronger crop/code controls are available. OCRBench is a weak positive case because routed admission is one selected

task above direct answer and above unconditional OCR. SEED-Bench-2-Plus is an abstention case because the router calls no tools and ties the direct path.

One task episode proceeds as follows. The agent receives the benchmark item and builds the observable task view. The direct-answer baseline produces a response without tool calls. The always-tool controls force a specific tool access pattern under the same family split. UtilityRouter instead computes its admission decision from the allowed task view and first-pass signal, then either answers directly or requests the selected tool observation before final answer synthesis. Only after the final prediction is made does the official or dataset-specific scorer assign the binary success value. This order is the non-oracle boundary: the route decision can depend on what the agent could observe before answering, but not on which condition later scores correctly.

The observed family-level routing deltas are interpreted with the main results.

The current implementation should be read as a diagnostic policy rather than an optimized controller. It is useful precisely because the errors are legible: the router can abstain, call a tool, or choose a single tool family, and each choice can be compared against an explicit control. When MathVista crop/code controls beat UtilityRouter, the failure is not hidden inside an opaque aggregate; it points to a concrete admission error. When SEED-Bench-2-Plus ties direct answer while always-tool controls are lower, the router’s abstention behavior becomes a budget-preserving outcome rather than a positive accuracy result.

4 Benchmarks and Experimental Setup

Benchmark families. The final selected-slice evidence uses three real benchmark families. MathVista contributes visual mathematical reasoning tasks scored by exact match after official normalization. OCRBench contributes text-rich visual tasks scored by answer substring match. SEED-Bench-2-Plus contributes text-rich visual multiple-choice tasks scored by official multiple-choice accuracy. Each final family uses 30 selected official tasks. The slice size is intentionally visible in every table caption and discussion paragraph. The experiments are full executions of the selected terminal matrix, not benchmark construction alone, but they are not leaderboard-wide claims for the source

benchmarks. This lets the paper make a defensible statement about the observed utility regimes while keeping benchmark-wide generalization in the limitations. The family choice is also purposeful: MathVista stresses visual math and program-like reasoning, OCRBench stresses text extraction, and SEED-Bench-2-Plus stresses text-rich visual comprehension where direct recognition can already be strong. The three-family design also prevents a single tool prior from dominating the interpretation. If the final evidence came only from OCRBench, the weak OCRBench gain could be mistaken for a general OCR-routing result. If it came only from MathVista, the router would look like an under-admitting policy with no positive signal. If it came only from SEED-Bench-2-Plus, the method would reduce to an abstention diagnostic. Keeping all three families in one terminal matrix lets the draft report the mixed pattern directly. The design also clarifies what the terminal matrix is not measuring. It does not replay full web-agent benchmarks, operate desktop environments, or train a search policy; those settings require action histories, live page or operating-system state, and service dependencies beyond this answer-level matrix. Instead, the selected tasks are used as a controlled measurement surface for the earlier admission decision: should the agent answer directly, request a visual/text/tool observation, or abstain from a tool call? This boundary keeps the evidence aligned with the available rows and prevents frontier web-agent citations from being mistaken for executed claims.

Conditions. All three families include direct answering, always OCR, always crop/zoom, and UtilityRouter. MathVista additionally includes always code and a code-only router. OCRBench and SEED-Bench-2-Plus include an OCR-only router. The evaluated answer backend is the hosted gpt-5-mini multimodal backbone used through the same answer interface for every condition. Direct and forced-tool baselines use one final answer call per task; UtilityRouter uses one first-pass uncertainty call plus one final answer call, and tool-mediated conditions add the corresponding OCR, crop/zoom, or code observation before final answer synthesis. The baseline roles are designed to isolate different explanations. Direct answer tests whether the evaluated multimodal backend can solve the row without external observations. Always-tool controls test whether access alone is beneficial or whether it

creates cost and distraction. Single-tool routers test whether the family has a dominant tool path that the general router should discover. UtilityRouter is then interpreted relative to all three controls, not only relative to direct answering. This is why MathVista is a failure case even though UtilityRouter matches direct answer: the stronger crop/code controls show that a useful tool was available.

Metrics and budget. Scores are binary task success values averaged within a benchmark family and condition. The table reports Wilson confidence intervals from the generated analysis artifact. Paired comparisons use exact two-sided sign tests over non-tied task pairs, with all-tie cases reported as not statistically distinguishable. Tool-call and API-call counts are tracked as budget descriptors rather than optimized objectives in this first draft. All answer calls use deterministic decoding for task scoring; the paper reports mean answer-call and tool-call counts to make tied outcomes comparable to direct answering. The budget descriptors are intentionally simple in this draft. They are sufficient to show when a method ties direct answer while consuming more calls, but they are not yet a full latency or monetary-cost frontier. For example, UtilityRouter records two API calls per task in the aggregate artifacts because it uses a first-pass signal and then a final answer path, while direct answer records one. This makes tied accuracy outcomes less favorable for the router unless the tool trace provides another audit or reliability benefit. The paper therefore avoids saying that tied families are wins. This accounting also affects how the single-tool controls are interpreted. Always-tool rows are not proposed deployment policies; they are stress tests for the sign of a tool’s marginal utility under matched tasks. Single-tool routers are narrower policies that ask whether one tool family alone explains the useful cases. UtilityRouter is expected to be most valuable only when the best action varies across rows and when observable cues can identify that variation. The current results show that this expectation is only partly met.

5 Main Results

Table 1 summarizes the central comparison. On MathVista, UtilityRouter scores 0.667, exactly matching direct answer, while always crop/zoom and the code-only router score 0.767. On OCRBench, UtilityRouter scores 0.900, one task above direct answer and OCR-only routing at 0.867. On

Benchmark/source (split; metric)	Backend	Method role	Calls	Tools	Score
<i>MathVista / mathvista (30 selected tasks; exact match)</i>					
	gpt-5-mini	Direct answer	1.00	0.00	0.667
	gpt-5-mini	Always code	1.00	1.00	0.700
	gpt-5-mini	Always crop/zoom	1.00	1.00	0.767
	gpt-5-mini	Always OCR	1.00	1.00	0.667
	gpt-5-mini	Code-only router	1.00	1.00	0.767
	gpt-5-mini	UtilityRouter	2.00	0.47	0.667
<i>OCRBench / ocrbench (30 selected tasks; substring match)</i>					
	gpt-5-mini	Direct answer	1.00	0.00	0.867
	gpt-5-mini	Always crop/zoom	1.00	1.00	0.833
	gpt-5-mini	OCR-only router	1.00	0.77	0.867
	gpt-5-mini	Always OCR	1.00	1.00	0.833
	gpt-5-mini	UtilityRouter	2.00	0.77	0.900
<i>SEED-Bench-2-Plus / seedbench2plus (30 selected tasks; multiple-choice accuracy)</i>					
	gpt-5-mini	Direct answer	1.00	0.00	0.733
	gpt-5-mini	Always crop/zoom	1.00	1.00	0.667
	gpt-5-mini	OCR-only router	1.00	0.00	0.733
	gpt-5-mini	Always OCR	1.00	1.00	0.700
	gpt-5-mini	UtilityRouter	2.00	0.00	0.733

Table 1: Main selected-slice matrix with deterministic hosted gpt-5-mini decoding: UtilityRouter is 20/30 on MathVista, 27/30 on OCRBench, and 22/30 on SEED-Bench-2-Plus; direct answer is 20/30, 26/30, and 22/30 respectively. Calls are answer-model calls per task; Tools are mean tool observations per task.

SEED-Bench-2-Plus, UtilityRouter, direct answer, and OCR-only routing all score 0.733. This result is not evidence for universal routing superiority. It is evidence that the same routing rule interacts differently with different benchmark families.

The confidence intervals also discourage over-interpretation. MathVista direct answer and UtilityRouter share the same 0.488–0.808 Wilson interval. OCRBench UtilityRouter has a 0.744–0.965 interval, while direct answer has 0.703–0.947, so the one-task difference is not a stable separation. SEED-Bench-2-Plus direct answer and UtilityRouter again share the same 0.556–0.858 interval. The table is therefore best read as a regime map: MathVista exposes missed crop/code utility, OCRBench exposes a weak positive routing signal, and SEED-Bench-2-Plus exposes a largely abstaining tie.

Figure 1 plots the same family-level deltas against the direct-answer baseline, which makes the mixed regime visible before the detailed failure discussion.

The always-tool rows add another layer. On MathVista, always crop/zoom and code-only routing are both 0.100 above UtilityRouter, contradicting a blanket over-calling story. On OCRBench, always OCR and always crop/zoom are 0.033 below direct answer and 0.067 below UtilityRouter, suggesting that unconditional tool access is not the

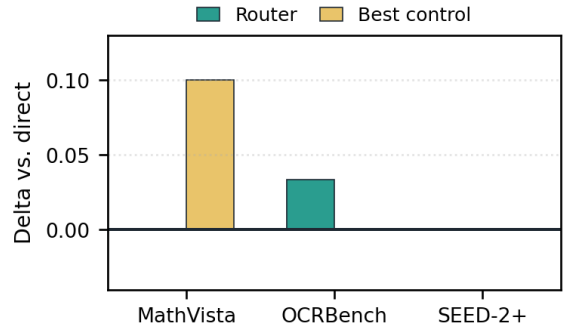


Figure 1: Router delta by family: UtilityRouter is +0.000 on MathVista, +0.033 on OCRBench, and +0.000 on SEED-Bench-2-Plus versus direct answer; the best non-direct control is +0.100 on MathVista, exposing the admission miss.

same as useful routing. On SEED-Bench-2-Plus, always OCR and always crop/zoom are below direct answer, while UtilityRouter makes no tool calls and ties direct answer. Together these rows support the narrower claim in the title of the analysis: tool utility is family-specific and route admission must be evaluated against explicit controls.

The paired tests reinforce the conservative interpretation. Against direct answer, MathVista and SEED-Bench-2-Plus are all-tie comparisons with 30 ties and $p=1.000$. OCRBench has one UtilityRouter win, no losses, and 29 ties, also $p=1.000$

under the two-sided sign test. Thus, the one-task OCRBench improvement is a useful diagnostic signal but not a statistically significant effect. The same paired comparison also shows why the MathVista failure should remain in the main body rather than only in limitations. Against always crop/zoom, UtilityRouter has one win, four losses, and 25 ties. Against the code-only router, it has no wins, three losses, and 27 ties. Neither comparison is significant, but both point in the same direction: the router is not admitting the strongest MathVista tool path often enough. This is the clearest evidence-backed failure mode in the current draft. The table also motivates why this draft does not collapse all families into a single macro-average. A macro-average would hide the fact that OCRBench is the only family with a router-over-direct selected-task gain, while MathVista contains the main tool-selection miss. For routing papers, that distinction is substantive: a policy that wins only by averaging one weak positive family with two tied families is not the same as a policy that reliably decides when tools help. The paper therefore treats the family-level table, paired comparison, and failure taxonomy as co-primary evidence, with the paired-test details kept in prose rather than isolated in a separate low-density float.

6 Analysis and Failure Cases

The largest failure mode is not over-calling everywhere; it is selecting the wrong tool family for a benchmark slice. MathVista shows this most clearly: always crop/zoom and the code-only router both beat UtilityRouter by 0.100 absolute score, while UtilityRouter ties direct answer. This pattern suggests that the observable heuristic under-admits the stronger visual/code path for some mathematical visual rows. There are two plausible explanations consistent with the saved rows. One is cue insufficiency: the observable pre-answer signals may not separate rows that need crop/zoom or code from rows that the base model can answer directly. The other is conservative thresholding: a training-free router can avoid bad tool calls but still fail to admit useful ones. The present artifacts do not distinguish these explanations conclusively, so the paper reports the measurable symptom and leaves threshold-learning or richer cue design as future work. This failure is also useful for method design because it rules out a common but unsupported rescue claim. The problem is not simply that tools are

noisy and should be avoided. On MathVista, some tool-mediated controls are better than direct answering on the selected slice. The error is therefore at the admission boundary: the current observable signals do not route enough rows toward the useful crop/code behavior. That is a narrower and more actionable diagnosis than saying that multimodal tools are globally helpful or globally harmful.

OCRBench shows the opposite shape. UtilityRouter is one selected task above direct answer, while always OCR and always crop/zoom trail direct answer by 0.033. This is the only final family where the router is higher than direct answer, but the paired test has only one non-tied task. The result should be read as an OCR-family routing signal to investigate, not as proof that tool routing generally improves text-rich visual tasks. The weak OCRBench signal is still useful because it rules out a trivial interpretation that all tool access hurts. The router calls tools on roughly 0.77 tasks per item on OCRBench and reaches 0.900, while always OCR calls a tool on every row and reaches 0.833. The gap between routed and unconditional OCR suggests that admission matters, even if the current selected-slice evidence is too small and tied-heavy for significance language. This is the paper’s strongest positive example of family-specific utility. The correct next experiment for this regime is scale, not stronger wording. A larger OCRBench slice would test whether the one selected-task gain persists and whether the admitted-tool subset is stable. Until then, OCRBench should be described as a weak positive signal and a useful case study for admission, not as a significant improvement.

SEED-Bench-2-Plus is mostly an abstention/tie case. UtilityRouter makes no tool calls in the selected slice and ties direct answer. Always OCR and always crop/zoom are lower than direct answer, which is consistent with the broader warning that tool access can add cost without improving accuracy. However, because the router abstains, this family supports non-oracle feasibility and budget discipline more than active tool-selection quality. This case is important for calibration. A router that calls no tools and ties direct answer should not be celebrated as improving accuracy, especially when it still uses the first-pass/final-answer protocol. At the same time, abstention is preferable to forced OCR or crop/zoom on this slice, where unconditional tool use is lower. The correct paper claim is therefore that the protocol can expose abstention regimes, not that abstention is always optimal.

This distinction matters for reviewer-facing calibration. Abstention can be a useful behavior when a direct multimodal backend is already strong, but a router that abstains everywhere would not be an orchestration method. The present SEED-Bench-2-Plus slice therefore supports the evaluation protocol and the cost-accounting story, while MathVista and OCRBench carry the active tool-selection evidence.

The generated failure taxonomy gives the same picture. MathVista has 26 router/direct ties and 4 rows where a non-router condition is better than UtilityRouter. OCRBench has 29 ties and 1 router win over direct answer. SEED-Bench-2-Plus has 29 ties and 1 row where a non-router condition is better than UtilityRouter. These counts justify a failure-analysis section, but not a broad performance claim.

Claim-level consequences. The executed rows support non-oracle protocol feasibility, family-specific tool-utility analysis, and the contrast between a weak OCRBench routing signal and a MathVista tool-selection failure. They do not support broad routing superiority, search/browser utility, or competitiveness with trained multimodal-search agents. The paper therefore treats search, browser, and trained-agent results as motivation for future extensions rather than as conclusions from the current matrix.

What would change the conclusion. The current conclusion would change only if additional executed evidence altered one of the three family regimes. A larger OCRBench run could show whether the one-task gain persists. A MathVista router with richer admissible cues could test whether the crop/code gap is a routing-feature problem or a structural limitation of this training-free policy. A provisioned search/browser family with the same direct, always-tool, single-tool, and router controls would be needed before any search or browsing utility claim can move from motivation to result. The same admission protocol would need to be extended before it could speak to long-horizon web or computer-control benchmarks. For those tasks, a route is not a single pre-answer action but a sequence of observation and interaction choices. An honest extension would therefore report trajectory-level controls, search/browser costs, and failure categories separately from the current answer-level OCR/crop/code controls. Until such rows are executed, the paper treats web-agent and

operating-system benchmarks as motivation for the routing problem rather than as evidence that UtilityRouter solves those domains.

7 Conclusion

This draft evaluates training-free multimodal tool routing as an auditable measurement problem. On a 480-row selected-slice matrix from three real benchmark families, UtilityRouter is non-oracle and exposes family-specific utility, but it is not a broad accuracy winner. The useful result is therefore a calibrated protocol: route decisions should be judged against direct, always-tool, and single-tool controls, and unsupported search/browser or trained-agent claims should remain out of scope until matching final evidence exists.

Limitations

The final matrix uses selected official slices of 30 tasks per family, so the reported numbers should be read as scoped evidence rather than benchmark-wide leaderboard claims. The paired tests contain many ties and no significant direct-answer comparison. The scoring pipeline combines official or dataset-specific normalization with stored row-level predictions, so reproducibility depends on preserving the selected task lists, predictions, scorer versions, and aggregate scripts with the release package. Search and browser claims are not supported by the terminal matrix, and no trained multimodal-search comparator is executed on the same split.

Ethical Considerations

Tool-routing papers can encourage overconfident deployment of autonomous agents if scoped measurements are reported as general autonomy. This draft therefore reports negative and tied outcomes directly, distinguishes final evidence from scoped-out search/browser dependencies, and avoids claims of trained-agent competitiveness. The evaluated benchmarks are public research benchmarks; any release package should preserve their licenses, attribution, and data-use restrictions.

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